

Asymptotic center–manifold for systems with parameter perturbations

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Asymptotic center-manifold approximations are constructed for systems that have been perturbed slightly away from the bifurcation point in the parameter space. This is done by extending the existing system to include parameter perturbations as dynamic variables, leading to an extended critical subspace and dimensionality of the center-manifold. The extension is done in a very general setting and it is shown that the consequences of such a system extension are non-trivial. The extended critical subspace has a certain structure associated with it, which may be deduced from the original system. The application of the center-manifold theorem to the extended system leads to a graph equation which rigorously valid in the vicinity of the bifurcation point (even in parameter space). The graph equation is approximated as an asymptotic power series in the amplitudes of the critical eigenmodes leading to general expressions for restricted resolvent solutions for the various terms in the power series at each order. A Ginzburg-Landau model near bifurcation is used to demonstrate the validity of the asymptotic center-manifold in the extended space.

I. PROBLEM DEFINITION

A. The Standard System

Consider a system with its state variable denoted by $\mathbf{U}$, consisting of l parameters given by the vector $\boldsymbol{\mu}$, and an evolution law given by

$$\frac{\partial \mathbf{U}}{\partial t} = \mathbf{f}(\mathbf{U}, \boldsymbol{\mu}). \quad (1)$$

The evolution law may be a finite set of ordinary differential equations or, an infinite dimensional partial differential equation as often arises for the space-time evolution of physical systems. Typically, physical systems will have real parameters $\boldsymbol{\mu}$ however, allowing for complex parameters presents no additional difficulty and we assume that $\boldsymbol{\mu} \in \mathbb{C}^l$. For a certain set of parameter values $\boldsymbol{\mu} = \boldsymbol{\mu}^b$, the system is assumed to have a fixed point for a state vector $\mathbf{U}^b$ satisfying

$$0 = \mathbf{f}(\mathbf{U}^b, \boldsymbol{\mu}^b). \quad (2)$$

The system may be rewritten in terms of the deviation from the fixed point solution $\mathbf{U} = \mathbf{U}^b + \mathbf{u}$ such that

$$\begin{aligned} \frac{\partial \mathbf{u}}{\partial t} = & \mathcal{L}_{\mathbf{U}}(\mathbf{u}) + \\ & [\mathbf{f}(\mathbf{U}^b + \mathbf{u}, \boldsymbol{\mu}^b) - \mathcal{L}_{\mathbf{U}}(\mathbf{u})], \end{aligned} \quad (3)$$

where, $\mathcal{L}_{\mathbf{U}}$ is the linearized operator at the fixed point, acting on $\mathbf{u}$, defined by

$$\mathcal{L}_{\mathbf{U}}(\mathbf{u}) = \left(\left. \frac{\partial \mathbf{f}}{\partial \mathbf{U}} \right|_{(\mathbf{U}^b, \boldsymbol{\mu}^b)} \right) \cdot \mathbf{u}. \quad (4)$$

The right hand side in equation (3) has been divided into the linear and non-linear contributions with $\mathcal{L}_{\mathbf{U}}(\mathbf{u})$

representing the linear part and the remaining terms in the square brackets representing the non-linear contributions.

Furthermore, we assume that $\boldsymbol{\mu}^b$ represents the parameters of the system at a bifurcation point. It is generically assumed that the system undergoes a multiple co-dimension bifurcation at $\boldsymbol{\mu}^b$ and that the linearized operator $\mathcal{L}_{\mathbf{U}}$ has a discrete set of n critical eigenvalues λ^C , with zero growth rate ($\Re(\lambda^C) = 0$). This includes both purely real critical eigenvalues with $\Re(\lambda^C) = 0$, $\Im(\lambda^C) = 0$, and purely imaginary critical eigenvalues with $\Re(\lambda^C) = 0$, $\Im(\lambda^C) \neq 0$. For real systems complex eigenvalues occur in conjugate pairs. Accordingly, the system has n critical eigenpairs $(\lambda_i^C, \mathbf{v}_i^C)$, $i = 1 \dots n$, where $\mathbf{v}_i^C$ represents the eigenvector corresponding to λ_i^C .

The adjoint of the direct linearized operator $\mathcal{L}_{\mathbf{U}}$ is denoted as $\mathcal{L}_{\mathbf{U}}^\dagger$ which also has n critical eigenpairs $(\lambda_i^{C*}, \mathbf{w}_i^C)$. The scaling of the eigenvectors of the direct and adjoint operators may be defined such that they satisfy the biorthogonality condition

$$\langle \mathbf{w}_i^C, \mathbf{v}_j^C \rangle = \mathbf{w}_i^{C\dagger} \mathbf{v}_j^C = \delta_{i,j}, \quad (5)$$

where $\langle \cdot, \cdot \rangle$ is the standard inner-product and $\delta_{i,j}$ is the Kronecker-delta function. The superscript $\dagger$ is used here to indicate a complex-conjugated transpose of vectors and matrix operators. For scalars complex-conjugation is indicated by a $*$ superscript. The superscript H which is also often used for the same purpose is reserved for objects referring to the harmonic forcings. One may define a matrix $\mathbf{V}$, whose columns comprise of the n critical eigenvectors of $\mathcal{L}_{\mathbf{U}}$ and the matrix spans the invariant center subspace of the linear operator $\mathcal{L}_{\mathbf{U}}$. The corresponding matrix for the adjoint eigenvectors is denoted $\mathbf{W}$. Due to the chosen scaling the matrices satisfy the following relations,

$$\mathbf{W}^\dagger \mathbf{V} = \mathbf{I}_n \quad (6)$$

where, $\mathbf{I}_n$ represents an identity matrix of order $n \times n$. The projection operator into the stable subspace of $\mathcal{L}_{\mathbf{U}}$

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is defined by

$$\mathbf{Q} = \mathbf{I} - \mathbf{V}\mathbf{W}^\dagger. \quad (7)$$

Additionally, $\mathbf{\Lambda}_C$ is defined as an $n \times n$ diagonal matrix consisting of the critical eigenvalues of $\mathcal{L}_U$, with the implicit assumption of diagonalizability of $\mathcal{L}_U$ within this subspace. The ordering of the eigenvalues in $\mathbf{\Lambda}_C$ and eigenvectors in $\mathbf{V}$ is consistent with each other such that $\mathcal{L}_U \mathbf{V} = \mathbf{V} \mathbf{\Lambda}_C$.

B. The Extended System

The same system is now considered for the case of parameter perturbation so that the evolution dynamics are now evaluated at parameter values of $\boldsymbol{\mu} = \boldsymbol{\mu}^b + \boldsymbol{\nu}$, with $\boldsymbol{\nu}$ being the vector of parameter perturbations. In addition, the system is forced with small-amplitude harmonic forcing of the type

$$\mathbf{f}_H(t) = \sum_{i=1}^h \mathbf{f}_i \theta_i(t); \quad \theta_i = \epsilon_i e^{\lambda_i^H t}, \quad (8)$$

where, the various θ_i are treated as the dynamic variable, λ_i^H is purely imaginary, with the imaginary part representing the angular frequency of the forcing. $\mathbf{f}_i$ represents the shape of the i^{th} harmonic forcing and ϵ_i is a scale factor. Constant forcing terms are readily incorporated when a particular (or multiple) $\lambda_i^H = 0$. The vector of the various θ_i is denoted as $\boldsymbol{\theta} = [\theta_1; \theta_2; \dots; \theta_h]$, and the vector of angular frequencies is denoted as $\boldsymbol{\lambda}^H = [\lambda_1^H; \lambda_2^H; \dots; \lambda_h^H]$. The resulting evolution equation is

$$\frac{\partial \mathbf{u}}{\partial t} = \mathbf{f}(\mathbf{U}^b + \mathbf{u}, \boldsymbol{\mu}^b + \boldsymbol{\nu}) + \sum_{i=1}^h \mathbf{f}_i \theta_i.$$

The diagonal matrix for the harmonic angular frequencies is represented by $\mathbf{\Lambda}_H = \text{diag}(\boldsymbol{\lambda}^H)$. Similarly, $\boldsymbol{\lambda}^P = [\lambda_1^P; \lambda_2^P; \dots; \lambda_l^P]$ is defined as a vector of length l , with all $\lambda_i^P = 0$ identically, for $i \in 1 \dots l$. These may be thought of parameter eigenvalues of parameter angular frequencies depending on the point of view. In either case, these vanish identically. $\mathbf{\Lambda}_P = \text{diag}(\boldsymbol{\lambda}^P)$ is defined as an $l \times l$ null matrix, to represent the trivial linear evolution matrix for the parameter variables $\boldsymbol{\nu}$. The extended system including the evolution equations for the newly introduced variables $\boldsymbol{\nu}$ and $\boldsymbol{\theta}$, henceforth referred to as the *extended variables*, is given by,

$$\frac{\partial \mathbf{u}}{\partial t} = \mathcal{L}_U(\mathbf{u}) + \mathcal{L}_\mu(\boldsymbol{\nu}) + \mathcal{L}_\theta(\boldsymbol{\theta}) + \mathbf{f}_{NL}(\mathbf{u}, \boldsymbol{\nu}) \quad (9a)$$

$$\frac{d\boldsymbol{\nu}}{dt} = \mathbf{\Lambda}_P \boldsymbol{\nu} \quad (9b)$$

$$\frac{d\boldsymbol{\theta}}{dt} = \mathbf{\Lambda}_H \boldsymbol{\theta}, \quad (9c)$$

Here, as done earlier, the evolution equations are split into the linear and non-linear contributions. $\mathcal{L}_U$ is the

same linearized operator as defined earlier in equation 4. $\mathcal{L}_\mu$ is the linearized operator acting on the newly introduced extended variable $\boldsymbol{\nu}$, defined as

$$\mathcal{L}_\mu(\boldsymbol{\nu}) = \left(\frac{\partial \mathbf{f}}{\partial \boldsymbol{\mu}} \bigg|_{(\mathbf{U}^b, \boldsymbol{\mu}^b)} \right) \cdot \boldsymbol{\nu}. \quad (10)$$

Similarly, $\mathcal{L}_\theta$ is the linearized operator acting on $\boldsymbol{\theta}$, defined as,

$$\mathcal{L}_\theta(\boldsymbol{\theta}) = \left(\frac{\partial \mathbf{f}_H}{\partial \boldsymbol{\theta}} \right) \cdot \boldsymbol{\theta}. \quad (11)$$

This is the same expression as in equation (8), just written in a manner consistent with the other terms. Finally, $\mathbf{f}_{NL}(\mathbf{u}, \boldsymbol{\nu})$ represents the non-linear terms of the modified system given by

$$\mathbf{f}_{NL}(\mathbf{u}, \boldsymbol{\nu}) = \mathbf{f}(\mathbf{U}^b + \mathbf{u}, \boldsymbol{\mu}^b + \boldsymbol{\nu}) - \mathcal{L}_U(\mathbf{u}) - \mathcal{L}_\mu(\boldsymbol{\nu}), \quad (12)$$

where, the dependence on $\mathbf{U}^b$ and $\boldsymbol{\mu}^b$ has been suppressed for notational convenience.

An overhead hat notation, $\hat{\cdot}$, is used to refer to variables and operators associated with the extend space, so that the new state vector, non-linear term and the linearized direct and adjoint operators are denoted by $\hat{\mathbf{u}}$, $\hat{\mathbf{f}}_{NL}$, $\hat{\mathcal{L}}$ and $\hat{\mathcal{L}}^\dagger$, that are defined as,

$$\hat{\mathbf{u}} = \begin{Bmatrix} \mathbf{u} \\ \boldsymbol{\nu} \\ \boldsymbol{\theta} \end{Bmatrix}; \quad \hat{\mathbf{f}}_{NL} = \begin{Bmatrix} \mathbf{f}_{NL} \\ \mathbf{0} \\ \mathbf{0} \end{Bmatrix}, \quad (13)$$

$$\hat{\mathcal{L}}(\hat{\mathbf{u}}) = \begin{bmatrix} \mathcal{L}_U & \mathcal{L}_\mu & \mathcal{L}_\theta \\ \mathbf{0} & \mathbf{\Lambda}_P & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Lambda}_H \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \boldsymbol{\nu} \\ \boldsymbol{\theta} \end{Bmatrix}, \quad (14)$$

$$\hat{\mathcal{L}}^\dagger(\hat{\mathbf{u}}^\dagger) = \begin{bmatrix} \mathcal{L}_U^\dagger & \mathbf{0} & \mathbf{0} \\ \mathcal{L}_\mu^\dagger & \mathbf{\Lambda}_P^\dagger & \mathbf{0} \\ \mathcal{L}_\theta^\dagger & \mathbf{0} & \mathbf{\Lambda}_H^\dagger \end{bmatrix} \begin{Bmatrix} \mathbf{u}^\dagger \\ \boldsymbol{\nu}^\dagger \\ \boldsymbol{\theta}^\dagger \end{Bmatrix}. \quad (15)$$

The full system may then be represented compactly as

$$\frac{\partial \hat{\mathbf{u}}}{\partial t} = \hat{\mathcal{L}}(\hat{\mathbf{u}}) + \hat{\mathbf{f}}_{NL}. \quad (16)$$

The system defined by equation (9), or its compact form equation (16), with additional dynamical equations for the extended variables will be referred to as the *extended system*.

The extended system has a larger critical subspace as compared to the original system due to the $l + h$ additional variables introduced. While the extension of the system seems almost trivial, the eigenvector structure in the additional critical dimensions however is not. Nevertheless, the extended eigenspace structure can be deduced directly from the properties of the original system. There are however some subtleties involved when the system extension introduces eigenvalues that are resonant with the original linear system $\mathcal{L}_U$. The system extension is therefore deduced for the resonant and the non-resonant cases separately.

1. The Non-resonant case

First, consider the case when $\mathcal{L}_U$ has no zero eigenvalues and the harmonic forcing is not resonant with $\mathcal{L}_U$.

Observe that if $(\lambda, \mathbf{v})$ is an eigenpair of the original linearized operator $\mathcal{L}_U$, then $(\lambda, [\mathbf{v}; \mathbf{0}; \mathbf{0}])$ is an eigenpair of the extended linearized operator $\hat{\mathcal{L}}$, which may be verified by simple substitution. Hence all the original eigenpairs can be trivially extended to the larger system and therefore extensions of the original critical eigenpairs are given by $(\lambda_i^C, [\mathbf{v}_i^C; \mathbf{0}; \mathbf{0}])$, for $i \in 1, \dots, n$. The additional critical eigenpairs introduced due to system extension need to be deduced. If $\mathcal{L}_U$ is non-singular, meaning it does not contain any purely real critical eigenvalues, then this can be done in a straightforward manner. On the other hand if $\mathcal{L}_U$ is singular, the deduction is more subtle. Two sets of canonical unit vectors are defined such that,

$$\mathbf{e}_i^P = \underbrace{[0, 0, \dots, 0]_{i-1}}_{i-1}, 1, \underbrace{[0, 0, \dots, 0]_{l-i}}_{l-i}, \quad (17a)$$

$$\mathbf{e}_i^H = \underbrace{[0, 0, \dots, 0]_{i-1}}_{i-1}, 1, \underbrace{[0, 0, \dots, 0]_{h-i}}_{h-i}. \quad (17b)$$

The l eigenpairs generated by the system extension due to ν have the structure of the type $(\lambda_i^P, [\mathbf{v}_i^P; \mathbf{e}_i^P; \mathbf{0}])$, which satisfy

$$(\lambda_i^P \mathbf{I} - \mathcal{L}_U) \mathbf{v}_i^P = \mathcal{L}_\mu(\mathbf{e}_i^P); \quad \lambda_i^P = 0 \quad (18)$$

$$\forall i \in 1, \dots, l.$$

This can be verified by simple substitution into the eigenvalue equation for $\hat{\mathcal{L}}$. These shall be referred to as *parameter modes*.

Similarly, the h eigenpairs generated by system extension due to θ have the structure of the type $(\lambda_i^H, [\mathbf{v}_i^H; \mathbf{0}; \mathbf{e}_i^H])$, which satisfy

$$(\lambda_i^H \mathbf{I} - \mathcal{L}_U) \mathbf{v}_i^H = \mathcal{L}_\theta(\mathbf{e}_i^H), \quad (19)$$

$$\forall i \in 1, \dots, h.$$

These will be referred to as *forcing modes*.

In the adjoint case, for any critical eigenpair $(\lambda_i^{C*}, \mathbf{w}_i^C)$ of the original adjoint operator $\mathcal{L}_U^\dagger$, the corresponding eigenpair of the extended operator $\hat{\mathcal{L}}^\dagger$ is given by $(\lambda_i^{C*}, [\mathbf{w}_i^C; \boldsymbol{\zeta}_i^P; \boldsymbol{\zeta}_i^H])$, with,

$$(\lambda_i^{C*} \mathbf{I} - \Lambda_P^\dagger) \boldsymbol{\zeta}_i^P = \mathcal{L}_\mu^\dagger \cdot \mathbf{w}_i^C, \quad (20a)$$

$$(\lambda_i^{C*} \mathbf{I} - \Lambda_H^\dagger) \boldsymbol{\zeta}_i^H = \mathcal{L}_\theta^\dagger \cdot \mathbf{w}_i^C, \quad (20b)$$

$\forall i \in 1, \dots, n$. In fact this extension is valid for any eigenmode of the original adjoint operator, not just the critical modes.

The adjoint eigenvectors that arise due to the system extension have a special structure. The l *adjoint parameter modes* are given by $(\lambda_i^{P*}, [\mathbf{0}; \mathbf{e}_i^P; \mathbf{0}])$, for $i \in 1, \dots, l$, and, the h *adjoint forcing modes* are given by $(\lambda_i^{H*}, [\mathbf{0}; \mathbf{0}; \mathbf{e}_i^H])$, for $i \in 1, \dots, h$. Hence the spectral properties of the extended system can be easily deduced for the non-resonant extension of the system.

2. The Resonant case

On the other hand, if the system extension is such that some of the eigenvalues introduced are in resonance with the original system, $\mathcal{L}_U$, then the deduction of the eigenstructure is more subtle. Here we assume that $\mathcal{L}_U$ is singular with one or more zero eigenvalues and that some of the angular frequencies of the harmonic forcings coincide with the oscillatory eigenvalues of $\mathcal{L}_U$.

Again, eigenpairs $(\lambda, \mathbf{v})$ of the original system $\mathcal{L}_U$ can be trivially extended to $(\lambda, [\mathbf{v}; \mathbf{0}; \mathbf{0}])$ which satisfy the eigenvalue equation for $\hat{\mathcal{L}}$. Hence the extensions of the original critical eigenpairs are given by $(\lambda_i^C, [\mathbf{v}_i^C; \mathbf{0}; \mathbf{0}])$, for $i \in 1, \dots, n$, which is identical to the non-resonant case.

For the i^{th} *extended parameter mode*, define a vector $\boldsymbol{\gamma}_i^P \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\boldsymbol{\gamma}_i^P)_j = \begin{cases} \mathbf{w}_j^{C\dagger} \mathcal{L}_\mu(\mathbf{e}_i^P), & \text{if } \lambda_i^P = \lambda_j^C \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

Then $(\lambda_i^P, [\mathbf{v}_i^P; \mathbf{e}_i^P; \mathbf{0}])$ forms a generalized eigenmode which satisfies the modified relation,

$$(\lambda_i^P \mathbf{I} - \mathcal{L}_U) \mathbf{v}_i^P = \mathcal{L}_\mu(\mathbf{e}_i^P) - \mathbf{V} \boldsymbol{\gamma}_i^P; \quad \forall i \in 1, \dots, l, \quad (22a)$$

$$\mathbf{w}_j^{C\dagger} \mathbf{v}_i^P = 0; \quad \text{if } \lambda_i^P = \lambda_j^C. \quad (22b)$$

It can be verified that $\hat{\mathbf{u}} = [\mathbf{v}_i^P; \mathbf{e}_i^P; \mathbf{0}]$ is indeed a generalized eigenvector by checking that $(\lambda_i^P \mathbf{I} - \hat{\mathcal{L}}) \hat{\mathbf{u}} \neq \mathbf{0}$, but $(\lambda_i^P \mathbf{I} - \hat{\mathcal{L}})^2 \hat{\mathbf{u}} = \mathbf{0}$, which is the condition to be satisfied for a generalized eigenvector.

Similarly, for the i^{th} *extended forcing mode*, define a vector $\boldsymbol{\gamma}_i^H \in \mathbb{C}^n$, where the j^{th} element of the vector is given by,

$$(\boldsymbol{\gamma}_i^H)_j = \begin{cases} \mathbf{w}_j^{C\dagger} \mathcal{L}_\theta(\mathbf{e}_i^H), & \text{if } \lambda_i^H = \lambda_j^C \\ 0, & \text{otherwise.} \end{cases} \quad (23)$$

Then $(\lambda_i^H, [\mathbf{v}_i^H; \mathbf{0}; \mathbf{e}_i^H])$ forms a generalized eigenmode which satisfies the modified relation,

$$(\lambda_i^H \mathbf{I} - \mathcal{L}_U) \mathbf{v}_i^H = \mathcal{L}_\theta(\mathbf{e}_i^H) - \mathbf{V} \boldsymbol{\gamma}_i^H; \quad \forall i \in 1, \dots, l, \quad (24a)$$

$$\mathbf{w}_j^{C\dagger} \mathbf{v}_i^H = 0; \quad \text{if } \lambda_i^H = \lambda_j^C. \quad (24b)$$

Again, It may be verified that for $\hat{\mathbf{u}} = [\mathbf{v}_i^H; \mathbf{0}; \mathbf{e}_i^H]$, the eigenvalue equation is not satisfied however, the generalized eigenvalue equation $(\lambda_i^H \mathbf{I} - \hat{\mathcal{L}})^2 \hat{\mathbf{u}} = \mathbf{0}$ is indeed satisfied.

For the adjoint eigenmodes, all non-resonant critical eigenmodes can be extended as has been done in equation (20). However, for critical adjoint eigenmodes that are resonant with the adjoint parameter and forcing modes, the eigenpairs are given by $(\lambda_i^{C*}, [\mathbf{w}_i^C; \mathbf{0}; \mathbf{0}])$, whenever $\lambda_i^{C*} = \lambda_j^{P*}$ or $\lambda_i^{C*} = \lambda_j^{H*}$.

, similar to the non-resonant case, any non-resonant eigenpair $(\lambda^*, \mathbf{w})$ of the original adjoint operator $\mathcal{L}_U^\dagger$, the

corresponding eigenpair of the extended operator $\widehat{\mathcal{L}}^\dagger$ is given by $(\lambda^*, [\mathbf{w}; \boldsymbol{\zeta}^P; \boldsymbol{\zeta}^H])$, with,

For all eigenpairs with $(\lambda^*, \mathbf{w}^\dagger)$ with $\lambda^* = 0$, the corresponding extended eigenpair is $(\lambda^*, \widehat{\mathbf{w}}^\dagger)$, with $\widehat{\mathbf{w}}^\dagger = [\mathbf{w}^\dagger; \mathbf{0}]$. As is the case for the direct problem, this corresponds to a generalized eigenvector which does not satisfy $\widehat{\mathcal{L}}^\dagger(\widehat{\mathbf{w}}^\dagger) = \mathbf{0}$, but rather satisfies $\widehat{\mathcal{L}}^\dagger(\widehat{\mathcal{L}}^\dagger(\widehat{\mathbf{w}}^\dagger)) = \mathbf{0}$. Finally, one obtains l additional eigenpairs with a special structure $(\lambda_i^*, [\mathbf{w}_i^\dagger; \boldsymbol{\zeta}_i^\dagger]) = (0, [\mathbf{0}; \mathbf{e}_i])$. That is, the adjoint eigenvector has only one non-zero element corresponding to the i^{th} parameter variable. These are referred to as the *adjoint parameter modes*.

Three pairs of matrices are defined for the different (generalized) eigenspaces and their corresponding projectors. The matrix $\widehat{\mathbf{V}}_I$ comprises of m columns that are the critical eigenvectors of $\widehat{\mathcal{L}}$ corresponding to the purely imaginary eigenvalues. The matrix $\widehat{\mathbf{W}}_I$ comprises of the corresponding eigenvectors of the adjoint extended operator $\widehat{\mathcal{L}}^\dagger$. The matrix $\widehat{\mathbf{V}}_R$ comprises of eigenvectors of $\widehat{\mathcal{L}}$ for the purely real critical eigenvalues, which are extensions of the purely real eigenvalues of the original operator $\mathcal{L}_U$. The matrix $\widehat{\mathbf{W}}_R$ is the counterpart of $\widehat{\mathbf{V}}_R$ for the adjoint problem. Both $\widehat{\mathbf{V}}_R$ and $\widehat{\mathbf{W}}_R$ have n columns. Finally, the matrix $\widehat{\mathbf{V}}_P$ comprises l columns of the parameter modes while $\widehat{\mathbf{W}}_P$ comprises of the adjoint parameter modes. In terms of the matrices defined earlier, the newly defined matrices have the following structure,

$$\widehat{\mathbf{V}}_I = \begin{bmatrix} \mathbf{V}_I \\ \mathbf{0} \end{bmatrix}; \quad \widehat{\mathbf{W}}_I = \begin{bmatrix} \mathbf{W}_I \\ \mathbf{Z} \end{bmatrix}; \quad (25a)$$

$$\widehat{\mathbf{V}}_R = \begin{bmatrix} \mathbf{V}_R \\ \mathbf{0} \end{bmatrix}; \quad \widehat{\mathbf{W}}_R = \begin{bmatrix} \mathbf{W}_R \\ \mathbf{0} \end{bmatrix}; \quad (25b)$$

$$\widehat{\mathbf{V}}_P = \begin{bmatrix} \mathbf{V}_P \\ \mathbf{I}_l \end{bmatrix}; \quad \widehat{\mathbf{W}}_P = \begin{bmatrix} \mathbf{0} \\ \mathbf{I}_l \end{bmatrix}. \quad (25c)$$

Obviously, $\widehat{\mathbf{V}}_P$ and $\widehat{\mathbf{W}}_P$ have no counter-parts in the original system. The three pairs of matrices arising from the direct and adjoint problems are bi-orthogonal leading

to the following relations,

$$\widehat{\mathbf{W}}_I^H \widehat{\mathbf{V}}_I = \mathbf{W}_I^H \mathbf{V}_I = \mathbf{I}_m, \quad (26a)$$

$$\widehat{\mathbf{W}}_R^H \widehat{\mathbf{V}}_R = \mathbf{W}_R^H \mathbf{V}_R = \mathbf{I}_n, \quad (26b)$$

$$\widehat{\mathbf{W}}_P^H \widehat{\mathbf{V}}_P = \mathbf{I}_l^H \mathbf{I}_l = \mathbf{I}_l, \quad (26c)$$

and all the other matrix products between adjoint and direct matrices vanishing. This is simply a restatement of the fact that the direct and adjoint eigenvector spaces of the extended system are biorthogonal. The extended critical subspaces are represented together in a single matrix with $(l + n + m)$ columns by $\widehat{\mathbf{V}} = [\widehat{\mathbf{V}}_P, \widehat{\mathbf{V}}_R, \widehat{\mathbf{V}}_I]$ and $\widehat{\mathbf{W}} = [\widehat{\mathbf{W}}_P, \widehat{\mathbf{W}}_R, \widehat{\mathbf{W}}_I]$. The projection operator that annihilates components in the critical subspace (or equivalently projects into the stable subspace) of $\widehat{\mathcal{L}}$ is defined by

$$\widehat{\mathbf{Q}} = \widehat{\mathbf{I}} - \widehat{\mathbf{V}} \widehat{\mathbf{W}}^H. \quad (27)$$

The $l \times l$ diagonal matrix containing the l zero eigenvalues due to system extension is defined as $\boldsymbol{\Lambda}_P$. Finally, a reduced matrix representing the oblique projection of the extended linear operator $\widehat{\mathcal{L}}$ onto the critical subspace is defined by,

$$\widehat{\mathbf{K}} = \widehat{\mathbf{W}}^H \widehat{\mathcal{L}} \widehat{\mathbf{V}} = \begin{bmatrix} \boldsymbol{\Lambda}_P & \mathbf{0} & \mathbf{0} \\ \boldsymbol{\Gamma} & \boldsymbol{\Lambda}_R & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Lambda}_I \end{bmatrix}, \quad (28)$$

which is a complex matrix of size $(l + n + m) \times (l + n + m)$. Both $\boldsymbol{\Lambda}_P$ and $\boldsymbol{\Lambda}_R$ are null matrices of the appropriate sizes. The diagonal elements of $\widehat{\mathbf{K}}$ represent the eigenvalues of the reduced system, that are the critical eigenvalues of the extended system, and may be represented by the vector $\widehat{\boldsymbol{\lambda}} = \text{diag}(\widehat{\mathbf{K}})$.

It is important to point out here that the extended variable $\boldsymbol{\nu}$ identically vanishes in the stable subspace of the extended linear system $\widehat{\mathcal{L}}$. This may be seen from the structure of the eigenmodes that were extended from the original system. The $\boldsymbol{\nu}$ component of all such modes identically vanishes. It is only the parameter modes that contribute to the extended variables and these modes are part of the critical subspace. This may also be understood from the dynamics perspective. The stable subspace represents the time-dependent decaying part of the linearized solution. If the decaying solution contributes to $\boldsymbol{\nu}$ then this would imply a time dependence of some (or all) of the parameters. This would contradict the trivial evolution equation for the parameter perturbations that was introduced.

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